1)

$$S = \frac{QK^T}{\sqrt{d_k}}$$

$P =$ softmax (S) (attention weights)

$A = PV$ (final attention output)

1) deriving $(\partial A/\partial V)$

$$A_{ij} = \sum_k P_{ik} V_{kj} \Rightarrow \frac{\partial A_{ij}}{\partial V_{ab}} = P_{ia} \delta_{jb}$$

matrix form $\nabla_V = P^T \nabla_A$

2) Softmax Jacobian

$$p_i = \frac{e^{s_i}}{\sum_k e^{s_k}}$$

(c1) when $i = j$, $\left(\frac{f}{g}\right)' = \left(\frac{f'g - gf'}{g^2}\right)$

$$\frac{\partial p_i}{\partial s_i} = \frac{e^{s_i}}{\sum_k e^{s_k}} - \left(\frac{e^{s_i}}{\sum_k e^{s_k}}\right)^2$$

$$\frac{\partial p_i}{\partial s_i} = p_i(1 - p_i)$$

(c2) when $i \neq j$, $\frac{\partial p_i}{\partial s_j} = (-p_i p_j)$

$$\frac{\partial p_i}{\partial s_j} = p_i(\delta_{ij} - p_j)$$

using the Kroneckar delta, we get $\frac{\partial p_i}{\partial s_j} = (-p_i p_j)$

$\left[\delta_{ij} = 1 \text{ for } i = j,\ 0 \text{ for } i \neq j\right]$ & $\frac{\partial p_i}{\partial s_j} = p_i(\delta_{ij} - p_j)$

Page No.
Date

2)

3) derive $\left(\frac{\partial A}{\partial Q}\right) \rightarrow$

$A = PV$

$$\nabla_P = (\nabla A V^T)$$

$P = \text{softmax}(S)$

$$\nabla S_{ij} = \sum_K \nabla P_{ik} \left(\frac{\partial P_{ik}}{\partial S_{ij}}\right)$$

$$\nabla S_{ij} = \left(\nabla P_{ij} P_{ij} - P_{ij} \sum_K \nabla P_{ik} P_{ik}\right)$$

$$\nabla S = P \odot (\nabla_P - \text{row-sums}(\nabla_P \odot P))$$

$$S = QK^T / \sqrt{dk}$$

$$\nabla_Q = \frac{\partial S}{\partial Q}(\nabla_S)$$

$$\nabla_Q = \frac{1}{\sqrt{dk}} [P \odot ((\nabla A V^T) - \text{row-sums}((\nabla A V^T) \odot P))]K$$

4) why the gradient vanishes and role of $\sqrt{dk}$?

When the input logits (S) to softmax func. have very large magnitudes, softmax distribution becomes incredibly sharp & one prob p_i gets pushed to 1 while others close to 0.

So, the Jacobian approaches zero in most cases thus gradient vanishes & network stops learning.

When we take dot product of two vectors Q & K of length dk, variance of resulting scalar grows

3)

linearly with d_k.

For large embedding dimensions, this naturally produces logits with massive magnitudes, pushing the softmax into those flat, zero gradient regions.